Machine Learning Classification of RNA Nucleobase Conformation// within Secondary Motifs

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Abstract

RNA structure determines molecular function; nucleobase conformation, in particular, is linked to active sites in RNA. Nucleobase conformation can be difficult to determine using atomic structure techniques. While many tools exist for three-dimensional structure prediction, none focus on structure within secondary motifs when some 3D characteristics are known. This paper explores various classification models for prediction RNA nucleobase conformation using primary and tertiary structure data within RNA secondary motifs.

1. Introduction

1.1. Background

RNA function is determined by its structure – the sequence of RNA nucleotides, how it folds, and how it interacts through intra- and intermolecular interactions. RNA structure is often split into the following levels:

- Primary structure is the sequence of nucleotide bases
- Secondary structure is the pattern of basic intramolecular interactions (such as loops, bulges, and pseudoknots)
- Tertiary structure: three dimensional folding patters, such as helices, sheets, and stacking.

RNA structure motifs are defined as recurrent structural elements containing multiple intramolecular RNA-RNA interactions, as observed in anatomic-resolution RNA structures (Leontis 2016). Figure 1 shows common secondary structure motifs.

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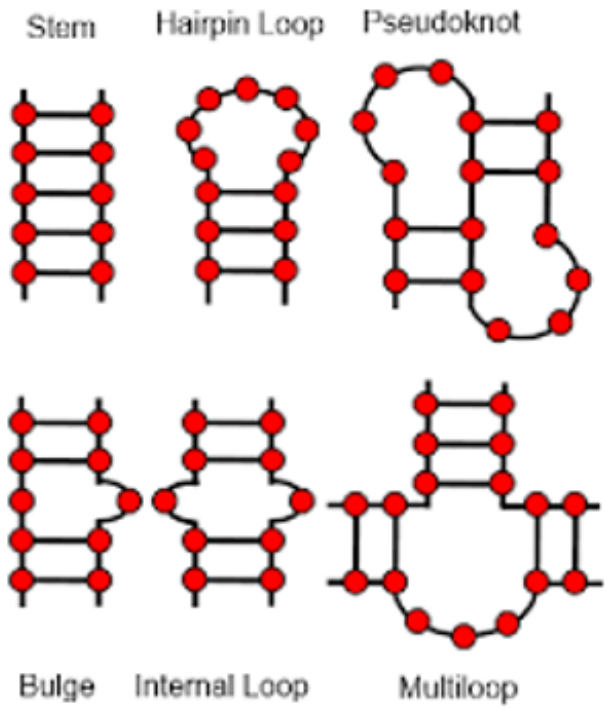

Figure 1. RNA secondary structure motifs. Motifs used in this project are Hairpin Loops, Bulges, and Internal Loops (symmetric and asymmetric).

Within the secondary and tertiary structure levels, there are several well-characterized types of three-dimensional structure features. Bases with multiple hydrogen bonds connecting them form base pairs, which can be classified into bae pair types, such as Watson-Crick, Wobble, Hoogsteen, reverse Hoogsteen and others. Nucleobases can also form single hydrogen bonds with other baes, or they can stack together. The way in which the sugar in a nucleotide is bent can be grouped into various exo- and endo- sugar puckers. These three-dimensional features are shown in Figure 2. Lastly, the angle of the nucleobase in relation to the sugar determines if the base is in the anti or syn conformation, as shown in Figure 3.

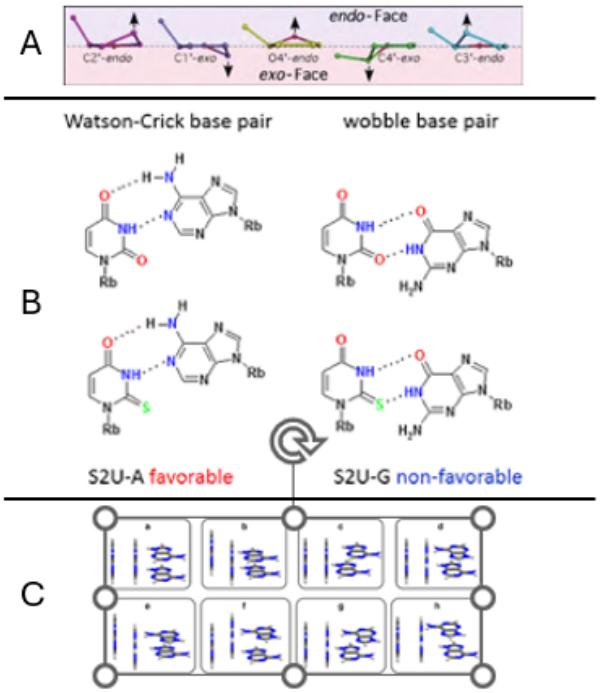

Figure 2. Tertiary structure features. (A) Sugar pucker angle and classification (endo- and exo-) (Ho, 2011) (B) Base pair categories (Duechler 2016) (C) Base stacking (Marek, 2019)

1.2. Literature Review

RNA structure prediction has been a large part of the application of machine learning to genetic material. There are now many programs which take primary or secondary structural input in order to predict the structure at higher levels. Programs such as SQUARNA, Pknots, and RNAfold predict secondary motifs from the primary sequence (Bohdan, 2024; Reeder, 2007; Hofacker, 2003). trRosettaRNA and RNA123 are two of the many programs which predict the entire three-dimensional molecule structure from the primary sequence (Wang, 2023; Eriksson, 2014). Some programs, such as RNAstructure and RNAcomposer also allow one to input secondary motifs, in addition to the primary sequence, to predict the three-dimensional structure (Bellaousov, 2013; Biersiada, 2016). RNAcomposer is a program which also allows for the incorporation of low-resolution structures with the primary structure to make more specific predictions (Biersiada, 2016). All of these programs focus on larger portions or the entirety of an RNA molecule, rather than looking within a single motif occurrence.

It has been well established that some clusters of RNA motifs, called sequence families, have a correlation between the primary sequence and the three-dimensional structure of the motif. The longest known family is the GNRA tetraloop (a hairpin with 4 unbonded bases in the loop) (Correll, 2003). Since then cluster analysis has been conducted in many different motifs and motif sizes (Saon, 2022, 2023). These studies generally focus on a single motif type and are used to characterize three-dimensional characteristics which make a family unique from other sequences of the same motif type.

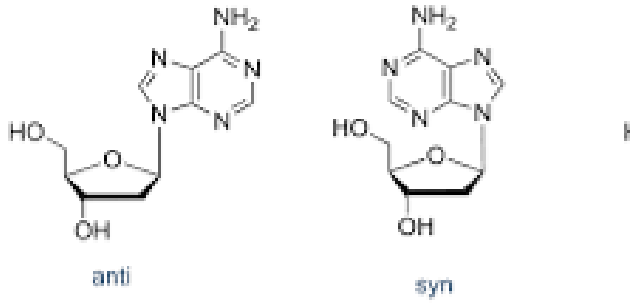

Figure 3. Anti/syn conformation of purine nucleobases (atdbio)

Previous studies have shown a link between the occurrence of nucleobases in the syn conformation and the base location in interactive portions of an RNA molecules, such as in activation sites. Additionally, the ability to detect nucleobase conformation can be difficult in NMR of larger molecules or in lower resolution X-ray crystallography. As more RNA molecules are studied and as NMR rises in popularity for measuring the three-dimensional RNA molecule, it is important to maintain the ability to determine nucleobase conformation.

1.3. Project Goal

The goal of the project is to develop and evaluate a series of classification models using machine learning techniques which use an input of primary, secondary, and three-dimensional structural data to predict nucleobase conformation across RNA secondary motifs.

2. Data

Data was gathered using the database RNA CoSSMos, a regularly maintained database of RNA structures which have been annotated for secondary structure motifs (Richardson, 2020). The raw data includes files from the Protein Data Bank for each RNA molecule (containing the base type, number and relative atomic coordinates for each nucleotide in the molecule) and an output from the CoSSMos site (with

the identification and location of the motifs within each .PDB file). A selection of data from each motif type and size was gathered; X-ray crystallography with resolution < 3 Angstroms and NMR outputs with a single structure were used. The full raw data output included over 200,000 motif instances.

3. Tools

- Protein Data Bank
- RNA CoSSMos database (Richardson, 2020)
- DSSR (Disecting the Spacial Structure of RNA)
- Python and relevant packages
  - Pandas
  - Numpy
  - SciKit Learn (for group shuffle split, LR, Gaussian NB, KNN, and ROC)
  - torch (for MLP)
  - imblearn (for oversampling and SMOTE)
  - Matplotlib

4. Methods

4.1. Preprocessing Data

4.1.1. Data Acquisition and Cleaning

Raw data from the protein databank was downloaded using the .PDB files identified in the RNA CoSSMos search output. The original .PDB files were clipped to include each motif and the two most adjacent closing base pairs. The quality of the clipped files was checked to ensure there were no empty files, missing bases, or duplicated atoms.

4.1.2. DSSR

After each motif is clipped and quality checked, the three-dimensional characteristics of each base are extracted. DSSR (disecting the spacial structure of RNA) is a program developed for the purpose of "analyzing and annotating RNA tertiary structures". This project used v2.3.2 - the basic package of version 2, last updated in 2021. The output from this step is a dataset for each of the following three-dimensional data: base pairing, hydrogen bonds, base stacking, and sugar pucker and base conformation.

4.1.3. Data Compilation

After running DSSR to extract tertiary features for each clipped PDB, the data was combined into a signle dataset. As a result of including the closing base pairs, each base of interest (e.g. any singular nucleotide within the motif) has at least 2 adjacent bases on each side. Each motif has a unique id sequence (pdb.id_motif.id); adding a position id number gives each base a unique identification. The output (y) is the conformation of the base of interest. The data which is the input is the primary, secondary, and other tertiary data of the base of interest and the four surrounding bases (input). A full list of the data for each sample can be found in Appendix A.

4.1.4. Data Cleaning 2

DSSR is unable to resolve nucleobase conformation for all samples; any sample with unknown conformation is removed from the database. Unresolved characteristics for other variables were changed to "NaN".

4.2. Splitting Data

The dataset was split into a validation set and 20-fold test and train sets. When splitting sets, the motif id sequence was used for a group shuffle split; this prevented data from different bases of the same motif from leaking across data splits. A seed was set to ensure replicability of data splits across models. Data from 500 motifs was partitioned out to use as the validation set. The remaining data was split into test and train sets using 20-fold LGOCV. The intersect of motif id sequences between sets was used to ensure there was no leakage of motifs; if the intersect was not an empty set, an error notification was printed.

4.3. Model Development

The data is an unbalanced dataset; less than 3% of the samples have bases in the syn conformation. With this in mind, each of the following models were trained using the original data, oversampled data, and SMOTE data.

4.3.1. Logistic Regression

The logistic regression models were built using the LogisticRegression function from Scikit learn. Models were built using the default LogLoss function, as well as L2 loss function and ElasticNet. Each of these were iterated over cutoff values (ranging from 1 to 8 with a step of 0.5) to determine the best cutoff. ROC was calculated and the curve was graphed. Evaluation metrics were accuracy, precision, recall, F1-score, and AUC and Pr-AUC scores. The best cutoff value was chosen as the cutoff with the highest average 20-fold F1-score. The model performance of different sampling methods and loss function combinations were then compared using the performance metrics listed above, with preference given to F1-score and accuracy.

4.3.2. Gaussian Naive Bayes

Gaussian Naive Bayes models were built using the GaussianNB package from Scikit learn. Both Naive Bayes and

complement Naive Bayes models were built. As with the Logistic regression, each model type iterated over various threshold cutoff values, using the F1-score to select the cut-off for each model. Accuracy, precision, recall, F1-score, and AUC and Pr-AUC scores were determined and the 20-fold averaged value was calculated.

4.3.3. K Nearest Neighbors

K-Nearest Neighbors models were built using the KNeighborsClassifier from Scikit learn. Various distance metrics were tested (the default Minkowski distance, Manhattan, and Hamming). Each of these was run using the default KNeighborsClassifier and with classification weighted by distance. Every model was iterated over several values of K (5,10,15,20). The best value of K was chosen using the F1-score. For each model (with its respective best value of K), the 20-fold average was calculated for accuracy, precision, recall, and F1-score.

4.3.4. Multi-Layer Perceptron

A Multi-layer Perceptron (MLP) model was chosen for this dataset because of its ability to use categorical data. ReLU was used for activation, and sigmoid function was applied for the output. A basic architectural diagram can be seen in Figure 6 (shown in MLP Results). The model was built using 5, 10, and 15 epochs, and each of these was iterated over threshold values. Precision, recall, accuracy, and F1-score (20-fold averages) were calculated for model evaluation.

4.4. Model Evaluation

The best performing model for each of the four classification model types was chosen based on the evaluation metrics listed above. That model was then retrained using the entirety of the test/train dataset. The four best models were then used to predict the classifications of the samples in the validation set. Precision, recall, accuracy and F1-score were calculated, and the macro average of each was used to compare model performance. F1-score and accuracy macro averages were prioritized for model evaluation.

5. Results

5.1. Logistic Regression

Based on the evaluation metrics, the best performing model had the following characteristics: SMOTE was used to handle imbalanced data. Loss was calculated using the LogLoss function. The threshold cutoff value was 0.65. Tabular comparison of logistic regression model performances can be found in the Appendix. Figure 4 shows the ROC curve, which shows that the model performs better than random chance.

Table 1. Evaluation metrics of Logistic Regression model; model uses LogLoss, SMOTE applied to test data, 0.65 cutoff threshold, values are 20-fold averages for 0/1 classes (0 = anti, 1 = syn conformations)

Metric	Train	Test
Precision	0.68/0.80	0.98/0.13
Recall	0.85/0.59	0.86/0.60
F1-score	0.75/0.68	0.86/0.60
Accuracy	0.72	0.85

Table 1 shows the evaluation metrics for the test and train sets. The accuracy is higher in the test set, which is a strong sign of a broken model, possibly due to data leakage. This phenomena was seen in most model iterations.

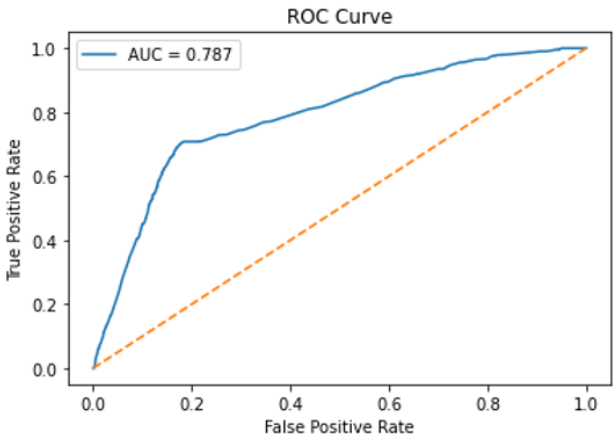

Figure 4. Logistic Regression: ROC Curve

5.2. Naive Bayes

Complement Naive Bayes performed far better than Naive Bayes in every iteration. SMOTE was used to handle the imbalance and the best cutoff ws 0.55. Tabular comparison of Naive Bayes model performances can be found in the Appendix. Figure 5 shows the ROC curve, which shows that this model is similar to the logistic regression classification in its AUC score and the shape of the ROC curve.

Table 2 shows the 20-fold average of the test and training evaluation metrics. The F1-score decreases drastically for the target class between train and test in this model. This model has the same issue as the Logistic Regression model, with the accuracy being higher in the test set.

5.3. K-Nearest Neighbors

Based on the evaluation metrics, K=15 was the highest performing number of neighbors. The classification decision

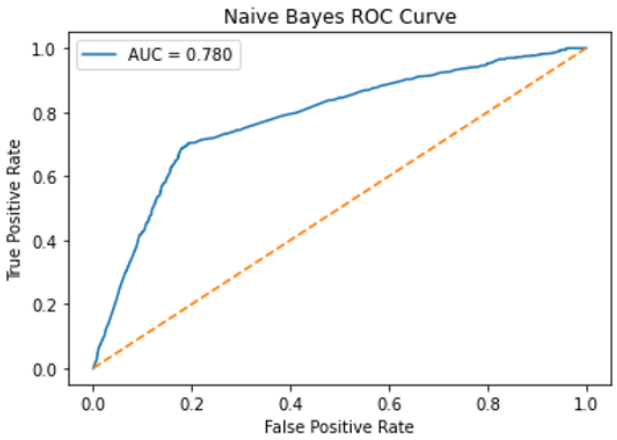

Figure 5. Naive Bayes ROC Curve

Table 2. Evaluation metrics of Naive Bayes model; model is complement gaussian naive bayes, SMOTE applied to test data, 0.55 cutoff threshold, values are 20-fold averages for 0/1 classes (0 = anti, 1 = syn conformations)

METRIC	TRAIN	TEST
PRECISION	0.69/0.77	0.998/0.11
RECALL	0.80/0.64	0.82/0.69
F1-SCORE	0.74/0.69	0.90/0.20
ACCURACY	0.72	0.82

was weighted by distance (as opposed to all 15 nearest neighbors contributing equally to the classification). The default Minkowski Distance was used. Tabular comparison of KNN model performances can be found in the Appendix

Table 3 Shows the evaluation metrics. Although this model has high accuracy, recall of the target class was very low in both the test and train sets. Further work is necessary to determine if other KNN models would misclassify fewer of the target class samples. Additionally, this model runs into the same issue of having higher accuracy in the test set.

5.4. Multi-Layer Perceptron

The MLP architecture used has 3 linear and 2 activation layers. Figure 6 shows a basic map of the architecture. ReLU is used for activation; the output is converted into a probability using application of sigmoid function. 15 epochs and a threshold of 0.5 were used in the best performing model.

Table 4 shows the evaluation metrics of the MLP model. This model has the highest recall of the target class, but the precision metrics and recall of the non target class indicate that there are a significant number of false positives. This is

Table 3. Evaluation metrics of K-Nearest Neighbors model; K=15, weight by distance, Minkowski Distance, values are 20-fold averages for 0/1 classes (0 = anti, 1 = syn conformations)

METRIC	TRAIN	TEST
PRECISION	0.66/0.78	0.97/0.17
RECALL	0.97/0.12	0.99/0.07
F1-SCORE	0.86/0.61	0.98/0.10
ACCURACY	0.73	0.96

Table 4. Evaluation metrics of Multi-Layer Perceptron model; threshold = 0.5, ReLU activation, epoch = 15, values are 20-fold averages for 0/1 classes (0 = anti, 1 = syn conformations)

METRIC	TRAIN	TEST
PRECISION	0.77/0.50	0.99/0.03
RECALL	0.01/1.00	0.01/1.00
F1-SCORE	0.03/0.67	0.03/0.06
ACCURACY	0.50	0.05

reflected in the the low F1-scores and accuracy.

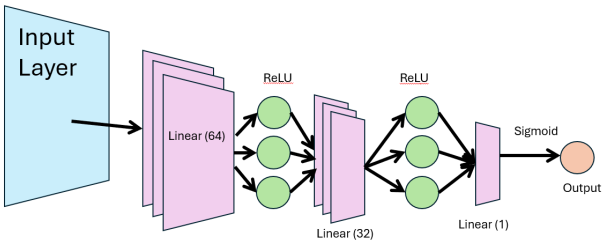

Figure 6. Multi-Layer Perceptron Architecture Diagram

5.5. Model Evaluation

6. Discussion

There were several challenges faced when building these classification models. The dataset is quite large (data was drawn from over 200,000 original motif clips and there are over 100 features); inclusion of oversampling methods, dense functions, KNN distance calculations, etc. have high computational costs. The models detailed in this paper, with the many iterations for sampling, parameter specifications, and thresholds, may require such techniques as dimension reduction in order to make computation costs feasible if any additional features or dense functions are included.

The dataset was very unbalanced. While oversampling and SMOTE techniques were used and compared in order to address the unbalanced output data, there were several other occurrences of unbalanced data in the input feature data

Table 5. Evaluation metrics of each model type using validation set; macro averages

METRIC	LR	CNB	KNN	MLP
PRECISION	0.56	0.55	0.57	0.51
RECALL	0.72	0.75	0.53	0.51
F1-SCORE	0.57	0.55	0.54	0.05
ACCURACY	0.86	0.80	0.96	0.05

which were not addressed due to time constraints. Even with the balancing techniques used on the training data, there were often large differences in the precision and recall of the target class (syn conformation) between the test and train sets.

Many models, in particular Naive Bayes and MLP, assume feature independence. Due to the nature of the data (as it is structural data of connected, adjacent molecules and sequential nucleotides), there is high correlation between some of the features, indicating the presence of dependence. A heat map of correlation of a subset of features can be found in the Appendix.

As discussed in the results, the discrepancy in accuracy poses a large challenge to the validity of the model. While the intersect of motif identification sequences can be used to ensure there is no leakage between test and train sets, it is possible that the original data includes .PDB files of different crystallographic or NMR measurements of the same RNA molecule or molecule type, which could have contributed to the accuracy issues.

6.1. Conclusion

Of the models previously described, Logistic Regression performed the best in terms of F1-score. However, with the data currently available and with the methods used, the models were low in precision and prone to overfitting across the board, and all but the MLP model faced challenges with accuracy discrepancies. Much future work is needed to determine the feasibility of applying classification models to the prediction of the nucleobase conformation.

6.2. Future Steps

The most immediate step is to address the accuracy issues; using regularization may resolve this.

Including more features (such as expanding the number of adjacent bases included for each base of interest, the location of the motif within the RNA molecule, and the type of RNA molecule) may improve the performance of the model. Additionally, prediction of nucleobase angle, rather than a binary classification, may improve the nuance of the model and its applications.

Table 6. Classification accuracies for naive Bayes and flexible Bayes on various data sets.

DATA SET	NAIVE	FLEXIBLE	BETTER?
BREAST	95.9± 0.2	96.7± 0.2	√
CLEVELAND	83.3± 0.6	80.0± 0.6	×
GLASS2	61.9± 1.4	83.8± 0.7	√
CREDIT	74.8± 0.5	78.3± 0.6	
HORSE	73.3± 0.9	69.7± 1.0	×
META	67.1± 0.6	76.5± 0.5	√
PIMA	75.1± 0.6	73.9± 0.5	
VEHICLE	44.9± 0.6	61.5± 0.4	√

This project only addressed one possible structure of an MLP deep learning model. Comparing this model to other configurations of MLP models may lead to higher performance.

Lastly, the deep learning model used (MLP) assumes feature independence. Other deep learning models (such as TabNet) which can handle related features should be explored.

Figure 7. Historical locations and number of accepted papers for International Machine Learning Conferences (ICML 1993 – ICML 2008) and International Workshops on Machine Learning (ML 1988 – ML 1992). At the time this figure was produced, the number of accepted papers for ICML 2008 was unknown and instead estimated.

Supplementary Material

Original data can be accessed using the database rnacossmos.com

Python scripts used can be accessed using the following GitHub link: https://github.com/klamp314/AML_Nucleobase_Classification. The scripts included are listed below.

- preprocess1.ipynb (Download, clip, clean motifs, run DSSR)
- preprocess2.ipynb (Compile dataset)
- lr_test.ipynb (Logistic regression model train/test)
- nb_test.ipynb (Naive Bayes model train/test)
- knn_test.ipynb (K-Nearest neighbors model train/test)
- mlp_test.ipynb (MLP model train/test)
- val_eval.ipynb (cross-model evaluation using validation dataset)

Appendices can also be accessed in the GitHub page.

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